Form-2

FINAL YEAR PROJECT

[PROPOSAL]

Project Title	AI-POWERED RESUME TAILORING & ATS OPTIMIZATION SYSTEM

DESCRIPTION

In today's competitive job market, over 75% of resumes are rejected by Applicant Tracking Systems (ATS) before reaching human recruiters. This project develops an AI-powered web application that intelligently tailors resumes to maximize ATS compatibility. The system accepts resumes in PDF, DOCX, and TXT formats, extracting key information including work experience, skills, education, and projects using advanced NLP techniques. Upon receiving a job description, the AI performs semantic matching to identify gaps, generates ATS-optimized keyword suggestions, and rewrites bullet points to align with job-specific terminology while preserving authenticity. The system includes a visual dashboard displaying match scores, keyword recommendations, and improvement suggestions. An explainable AI module provides transparency by highlighting modified sections with justifications. The final output is a formatted, ATS-friendly resume in DOCX and PDF formats. The application will be deployed as a responsive web platform accessible via desktop and mobile devices with secure user authentication. Technologies include FastAPI (backend), React.js (frontend), LangChain for LLM orchestration, PostgreSQL for data storage, and Groq's free API (Llama 3.2) for cost-effective text generation. Document parsing utilizes PyPDF2 and python-docx, while Sentence Transformers handle semantic similarity calculations. The entire system will be containerized using Docker for easy deployment and scalability.

PRIMARY GOALS

- PDF/DOCX Resume Parsing & Information Extraction
- Job Description Analysis & Keyword Extraction
- Semantic Matching Algorithm (Candidate vs. Job Requirements)
- AI-Powered Resume Rewriting & Tailoring (Using LLM)
- ATS Compatibility Scoring & Recommendations Dashboard
- Explainable AI Module (Highlighting Changes Made)
- Responsive Web Application (React + FastAPI)
- User Authentication & Resume Storage (PostgreSQL)

RESOURCES REQUIRED

Computer Lab, Internet, Python 3.10+, VS Code, GitHub, Groq API (Free Tier), HuggingFace Models (Sentence Transformers), Docker, PostgreSQL, React.js Libraries, PyPDF2, python-docx, LangChain

Student Name	Signature	Supervisor Signature